

Taming Beamer

This article introduces LaTeX and the Beamer extension class, which is ideal for creating professional-quality documents and presentations.



The OpenOffice suite, and even the new arrival, LibreOffice, have not matured enough as truly user-friendly professional document-processing tools. A simpler alternative for this is LaTeX (pronounced 'Lah-Tech' or 'Lay-Tech'), which is extremely popular in academic and scientific communities due to its rich functionality. Scientists prefer to create their research papers using LaTeX, publishing houses use LaTeX to type-set books, professors use LaTeX to create course handouts and presentations... LaTeX could even be used to design wedding cards and chess boards! Though it is slightly difficult to start learning LaTeX, once learned, its immense power will erase all the difficulties of creating a professional-quality document/presentation on a GNU/Linux system.

For the uninitiated, LaTeX is a highly efficient document preparation system developed by Leslie Lamport. It all started when the legendary author Donald Knuth was writing the second volume of *The Art of Computer Programming*. The lack of a good type-setting system led him to develop TeX and release it as free software. LaTeX can be thought of as a TeX extension intended for UNIX systems. Like TeX, LaTeX is also free software, released under the LaTeX Project Public License (LPL).

Why LaTeX?

One of the main advantages is that it helps type-set documents beautifully, especially technical documents:

- Mathematical formulas can easily be constructed using certain commands.
- Being a programming language, a user can easily modify its default behaviour.
- The availability of professionally crafted in-built layouts makes a document really look as if it has been printed.
- Documents in PS or PDF formats can be easily generated.

For more information, consider reading the well-written LaTeX Wikibook document that's available for free at <http://en.wikibooks.org/wiki/LaTeX>.

Classes and packages

Like the Linux mechanism of extending kernel features with modules, LaTeX provides classes and packages for extending behaviour. A class is used to modify a document style. Beamer is one of the most popular LaTeX classes for creating presentations. Prosper is another widely used presentation-making class. Apart from these, LaTeX also provides a default slide class to create slides.

Packages can be added and used inside a document class. A package is used to provide some additional functionality to an existing document. For example, if you need to insert a picture, you have to include the *graphics* package (`\usepackage{graphics}`). For a document, there can only be a single class, but many packages can be used inside a class. Some commonly used packages are: